

String Methods Demonstration in Java

1. Aim

To write and execute a Java program that demonstrates commonly used String methods such as `length()`, `trim()`, `charAt()`, `substring()`, case conversion, searching, comparison, replacement, and splitting.

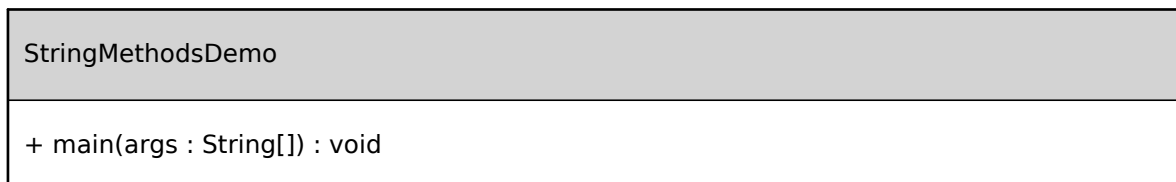
2. Description

This program demonstrates 15 commonly used Java String methods. It creates a String containing leading and trailing spaces and applies different methods to inspect, modify, search, compare, and split the string. `Arrays.toString()` is used to display the array produced by `split()`.

3. Algorithm

1. Start the program.
2. Import `java.util.Arrays`.
3. Create the string " Hello, Java World! ".
4. Apply `length()` and `trim()`.
5. Use `charAt()` and `substring()` to access and extract characters.
6. Use `toUpperCase()` and `toLowerCase()`.
7. Use `contains()`, `indexOf()` and `lastIndexOf()` for searching.
8. Use `replace()`, `startsWith()`, and `endsWith()`.
9. Use `equals()` and `equalsIgnoreCase()` for comparison.
10. Use `split()` to divide the string into words.
11. Display all results and end the program.

4. UML Diagram



5. Source Code with Comments

```
package core_java;
import java.util.Arrays;

public class StringMethodsDemo {
    public static void main(String[] args) {

        String str = " Hello, Java World! ";
        System.out.println("Original String: '" + str + "'\n");

        // 1. length() - Returns the number of characters
        int length = str.length();
        System.out.println("1. length(): " + length);

        // 2. trim() - Removes leading and trailing whitespace
        String trimmed = str.trim();
        System.out.println("2. trim(): '" + trimmed + "'");

        // 3. charAt(index) - Returns the character at a specific position
        char character = trimmed.charAt(7);
        System.out.println("3. charAt(7): " + character);

        // 4. substring(beginIndex, endIndex) - Extracts a portion
        String sub = trimmed.substring(7, 11);
        System.out.println("4. substring(7, 11): " + sub);

        // 5. toUpperCase() - Converts all characters to uppercase
        String upper = trimmed.toUpperCase();
        System.out.println("5. toUpperCase(): " + upper);

        // 6. toLowerCase() - Converts all characters to lowercase
```

5. Source Code with Comments (continued)

```
String lower = trimmed.toLowerCase();
System.out.println("6. toLowerCase(): " + lower);

// 7. contains(sequence) - Checks for a specific sequence
boolean hasJava = trimmed.contains("Java");
System.out.println("7. contains(\"Java\"): " + hasJava);

// 8. replace(old, new) - Replaces characters or substrings
String replaced = trimmed.replace("World", "Universe");
System.out.println("8. replace(): " + replaced);

// 9. indexOf(str) - Finds first occurrence of a substring
int firstIndex = trimmed.indexOf("a");
System.out.println("9. indexOf(\"a\"): " + firstIndex);

// 10. lastIndexOf(str) - Finds last occurrence of a substring
int lastIndex = trimmed.lastIndexOf("a");
System.out.println("10. lastIndexOf(\"a\"): " + lastIndex);

// 11. startsWith(prefix) - Checks the starting prefix
boolean starts = trimmed.startsWith("Hello");
System.out.println("11. startsWith(\"Hello\"): " + starts);

// 12. endsWith(suffix) - Checks the ending suffix
boolean ends = trimmed.endsWith("!");
System.out.println("12. endsWith(\"!\"): " + ends);

// 13. equals(anotherString) - Case-sensitive comparison
boolean isEqual = trimmed.equals("hello, java world!");
System.out.println("13. equals() [case-sensitive]: " + isEqual);

// 14. equalsIgnoreCase(anotherString) - Ignores case
boolean isEqualIgnore =
    trimmed.equalsIgnoreCase("hello, java world!");
System.out.println("14. equalsIgnoreCase(): " + isEqualIgnore);

// 15. split(regex) - Splits the string into an array
String[] words = trimmed.split(" ");
System.out.println("15. split(\" \"): " + Arrays.toString(words));
}
}
```

6. Compilation and Execution Commands

```
javac -d . StringMethodsDemo.java
java core_java.StringMethodsDemo
```

7. Output Screenshot

```
$ javac -d . StringMethodsDemo.java
$ java core_java.StringMethodsDemo

Original String: ' Hello, Java World! '

1. length(): 21
2. trim(): 'Hello, Java World!'
3. charAt(7): J
4. substring(7, 11): Java
5. toUpperCase(): HELLO, JAVA WORLD!
6. toLowerCase(): hello, java world!
7. contains("Java"): true
8. replace(): Hello, Java Universe!
9. indexOf("a"): 8
10. lastIndexOf("a"): 10
11. startsWith("Hello"): true
12. endsWith("!"): true
13. equals() [case-sensitive]: false
14. equalsIgnoreCase(): true
15. split(" "): [Hello,, Java, World!]
```

8. Result

The program was successfully compiled and executed. The output demonstrates the behavior and results of 15 commonly used Java String methods.